

DATA STRUCTURES								
II Year I Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS04	PCC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100
COURSE OBJECTIVES:								
1. Impart the basic concepts of data structures and algorithms. 2. Understand concepts linked lists and their applications. 3. Understand basic concepts about stacks, queues and their applications. 4. Understand basic concepts of trees, graphs and their applications. 5. Enable them to write algorithms for sorting and searching and hashing. 6. Use advanced data structures like B-Trees, AVL-trees etc., for efficient problem solving.								
COURSE OUTCOMES								
At the end of the course, student will be able to:								
1. Evaluate algorithms in terms of time and memory complexity. 2. Formulate new solutions for problems or improve existing code using data structures and algorithms. 3. Implement basic data structures such as arrays, linked lists, stacks and queues. 4. Solve problem involving graphs, trees and heaps 5. Apply Algorithms for solving problems like sorting, searching, and hashing. 6. Implement advanced data structures such as B-Trees, Red-Black, and AVL-Trees								
UNIT-I	INTRODUCTION TO DATA STRUCTURES							
Basic concepts- Algorithm Specification-Introduction, Recursive algorithms, Data Abstraction, Performance analysis- time complexity and space complexity, Asymptotic Notation-Big O, Omega and Theta notations, Introduction to Linear and Non Linear data structures-Singly Linked Lists- Operations-Insertion, Deletion, Concatenating singly linked lists, Circularlylinked lists-Operations for Circularly linked lists, Doubly Linked Lists- Operations- Insertion,Deletion. Representation of single, two dimensional arrays, sparse matrices-array and linkedrepresentations.								
UNIT-II	STACKS AND QUEUES							
Stacks-Stack ADT, definition, operations, array and linked implementations in C, applications-infix topostfix conversion, Postfix expression evaluation, recursion implementation, Queues-Queue ADT,definition and operations ,array and linked Implementations in C, Circular queues-Insertion anddeletion operations, Dequeue (Double ended queue)ADT, array and linked implementations in C.								
UNIT-III	TREES AND GRAPHS							
Trees – Terminology, Representation of Trees, Binary tree ADT, Properties of Binary Trees,Binary Tree Representations-array and linked representations, Binary Tree traversals, Threadedbinary trees, Max Priority Queue-ADT-implementation-Max Heap-Definition, Insertion into aMax Heap, Deletion from a Max Heap. Graphs , Introduction, Definition, Terminology, GraphADT, Graph Representations-Adjacency matrix, Adjacency lists, Graph traversals- DFS andBFS.								
UNIT-IV	SEARCHING AND SORTING							
Searching- Linear Search, Binary Search, Static Hashing-Introduction, hash tables, hashfunctions, Overflow Handling. Sorting-Insertion Sort, Selection Sort, Radix Sort, Quick sort, Merge Sort, Heap Sort, Comparison of Sorting methods.								
UNIT-V	BINARY SEARCH TREES							
Search Trees-Binary Search Trees, Definition, Operations- Searching, Insertion and Deletion,AVL Trees-Definition and Examples, Insertion into an AVL Tree ,B-Trees, Definition, B-Treeof order m, operations-Insertion and Searching, Introduction to Red-Black and SplayTrees(Elementary treatment-only Definitions and Examples), Comparison of Search Trees. Pattern matching algorithm- The Knuth-Morris-Pratt algorithm, Tries (examples only).								

TEXT BOOKS:

1. "Fundamentals of Data Structures", Illustrated Edition by Ellis Horowitz, Sartaj Sahni, Computer Science Press.
2. Fundamentals of Data structures in C, 2nd Edition, E.Horowitz, S.Sahni and SusanAnderson-Freed, Universities Press.

REFERENCE BOOKS:

1. Algorithms, Data Structures, and Problem Solving with C++", Illustrated Edition by Mark Allen Weiss, Addison-Wesley Publishing Company
2. "How to Solve it by Computer", 2nd Impression by R. G. Dromey, Pearson Education

WEB REFERENCES:

1. <https://hackr.io/tutorials/learn-data-structures-algorithms>
2. <https://www.geeksforgeeks.org/fundamentals-of-algorithms/>
3. <https://www.udemy.com/introduction-to-algorithms-and-data-structures-in-c/>
4. <https://leetcode.com>

E-TEXT BOOKS:

1. <http://www.freetchbooks.com/algorithm-analysis-and-design-t1030.html>
2. <http://www.freetchbooks.com/algorithmic-problem-solving-t373.html>
3. <http://www.freetchbooks.com/algorithms-and-data-structures-the-basic-toolbox-t871.html>

MOOC COURSE

1. <https://www.coursera.org/specializations/data-structures-algorithms>
2. https://onlinecourses.nptel.ac.in/noc16_cs06/preview